

Anomaly Detection in Subgraph Networks using Graph Neural Networks

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Abstract

Anomaly detection is an important problem in many application areas such as social network analysis, fraud detection, intrusion detection, and network security. The main goal of anomaly detection is to identify unusual patterns that do not follow normal behaviour. Traditional anomaly detection methods mainly focus on attribute data or textual data. However these methods do not effectively handle graph structured data. Graph data represent relationships between objects. In many real world systems anomalies appear as unusual patterns in the graph structure. Therefore subgraph anomaly detection becomes an important task. However only limited research studies focus on detecting anomalous subgraphs in graph networks. This paper proposes an unsupervised deep learning based method for subgraph anomaly detection. The proposed approach uses Graph Neural Networks (GNNs) to learn meaningful node embeddings from the graph structure and node attributes. After the embedding learning step a clustering algorithm groups nodes based on their embedding similarity. Each cluster forms a candidate subgraph. The method evaluates every subgraph using structural metrics and node attribute statistics. Threshold values are calculated for these metrics. A subgraph is marked as anomalous when its attribute values or structural characteristics deviate significantly from the threshold. The proposed method is evaluated using the Cora citation network dataset. Experimental results show that the method successfully identifies anomalous subgraphs and achieves high detection accuracy.

1 INTRODUCTION

Graphs are important data structures that represent relationships between entities. Many real world systems form graph structures. Examples include social networks, communication networks, biological networks,

and financial transaction networks. In these networks nodes represent entities and edges represent interactions between them. Subgraphs represent smaller structures inside a larger graph and help analyze localized behaviour in the network.

Detecting anomalous subgraphs is an important task in graph mining. An anomalous subgraph shows behaviour that differs from normal patterns in the network. These abnormal patterns may indicate fraudulent activities, network intrusions, malicious users, or abnormal information propagation. In social networks, anomalous subgraphs reveal coordinated fake accounts or unusual information spreading patterns [7].

Anomaly detection is a well studied problem in data mining and machine learning. Traditional anomaly detection methods focus mainly on tabular data or individual data points. However graph data contain complex structural relationships between nodes. Because of this relational structure, traditional methods do not capture abnormal patterns that exist inside graph structures.

Structural methods analyze properties such as node connectivity and graph density. These methods study how nodes connect inside a subgraph. An anomalous subgraph may show unusual connectivity patterns. It may also show abnormal density values when compared with normal subgraphs [1, 3]. Motif based methods study recurring patterns of nodes and edges in the graph. These patterns appear frequently in normal graphs. A deviation from these motifs indicates the presence of anomalous structures [16]. Attribute based methods analyze node and edge attributes such as profile information or activity patterns. Unusual attribute values indicate abnormal behaviour in the graph [21].

Despite these developments, subgraph anomaly detection remains a challenging problem. Graph data are often large scale and complex. The computational cost of analyzing all possible subgraphs is very high. It is also difficult to define appropriate anomaly thresholds and capture both structural and attribute information

together. These challenges become more significant in dynamic graphs where the network structure changes over time.

Recent advances in deep learning on graphs provide new opportunities to address these challenges. Graph Neural Networks (GNNs) learn low dimensional node representations that capture both structural information and node attributes. These learned representations improve the ability to detect abnormal patterns in graph data.

Subgraph anomaly detection has many important applications such as social network analysis, cybersecurity, bioinformatics, recommendation systems, and network monitoring. The detection of abnormal subgraph structures provides valuable insights into suspicious activities, hidden relationships, and abnormal behaviour patterns in complex networks.

2 METHODS AND MATERIALS

In our study, we present the Anomaly Encoder (AnomEn), a Graph Neural Network devised for detecting anomalies within graph-structured data. AnomEn operates as an encoder that processes high-dimensional adjacency and feature matrices and outputs a low-dimensional vector representation. The vector representations generated by AnomEn can then be integrated into machine learning models to classify subgraphs as either regular or anomalous. This offers a valuable tool for the detection of anomalies at the subgraph level, in addition to node and edge anomalies.

Drawing inspiration from the Graph Convolutional Network (GCN) [15], the AnomEn encoder aims to generate a robust latent representation of the graph data. Unlike the GCN model, which might be affected by anomalous neighboring nodes leading to potential false positives and negatives, AnomEn [8][9] mitigates these issues by striking a balance between the node's self-features and the attributes of its neighboring nodes. This process results in a more accurate representation of the graph data.

Each node's feature matrix constitutes its initial representation, $h_v^0 \leftarrow X_v$ for all $v \in V$. Messages are then exchanged among neighboring nodes, and every node receives messages from its neighbors as follows:

$$h_{N(v)}^k \leftarrow \text{AGGREGATE}(h_u^{k-1}, \forall u \in N(v)).$$

Subsequently, a node combines its features with those of its neighbors, granting its features more weight than the neighbors' features as denoted by:

$$h_v^k \leftarrow \sigma(W_k \cdot \text{COMBINE}(\beta h_v^{k-1}, (\beta - 1) h_{N(v)}^k)).$$

Here, $N(v)$ represents the set of neighbors of vertex v in a graph. In this way, each node has a latent representation that reflects its characteristics and is less influenced by anomalous neighbors.

2.1 Network Encoder

To map the input to node embeddings, two convolutional layers are used, which can be represented as:

$$\widetilde{Z}^E = \text{Relu} \left(D^{-1/2} (A(1 - \beta) + I\beta) D^{-1/2} (X)^T W^{(0)} + b^{(0)} \right) \quad (1)$$

$$Z^E = \text{Relu} \left(\widetilde{Z}^E D^{-1/2} (A(1 - \beta) + I\beta) D^{-1/2} W^{(1)} + b^{(1)} \right) \quad (2)$$

Here, $W^{(0)}$ is the weight matrix in the first layer, and $W^{(1)}$ is the weight matrix in the second layer.

During convolution, the aggregate of neighboring nodes' features is calculated by taking the dot product of the adjacency matrix and the feature matrix. An identity matrix, I , is added to the adjacency matrix to include a node's self-features. The terms $I \cdot \beta$ and $A \cdot (1 - \beta)$ signify the different weights assigned to the node's self-features and neighbors, respectively. The letter D stands for the diagonal matrix of A . The adjacency matrix is computed in the preprocessing phase as $(A \cdot (1 - \beta) + I \cdot \beta)$, where β denotes the weights assigned to the features of the node and its neighbors.

The aggregate is then given by the following equation:

$$\text{Aggregate} = (A \cdot (1 - \beta) + I \cdot \beta) \cdot \text{Features}$$

where:

- A is the adjacency matrix.
- β denotes the weights assigned to the features of the node and its neighbors.
- I is the identity matrix.
- Features represents the feature matrix.

After the training phase, the next step in our methodology involves extracting and normalizing the node embeddings. The objective of this extraction process is to obtain a meaningful numerical representation of each node that encapsulates its inherent characteristics and relationships with other nodes in the graph. Normalizing these embeddings ensures a standard scale, making subsequent computations more accurate and efficient.

Upon obtaining these normalized embeddings, we move on to the clustering phase. Clustering is an unsupervised machine learning technique which groups similar instances on the basis of certain features. In our context, these instances are the nodes represented by their embeddings, and the objective is to group similar nodes together into clusters. We utilize K-means clustering due to its simplicity and efficiency[6], making it ideal for our purpose.

The K-means clustering algorithm partitions the nodes into K distinct, non-overlapping subsets or clusters. The algorithm iteratively assigns each node to the cluster whose centroid is closest, where closeness is defined using Euclidean distance in the high-dimensional embedding space. These centroids are the central points (mean vectors) of the clusters.

To initialize the centroids, we use the K-means++ algorithm, a smart centroid initialization technique that can often result in better clustering outcomes than random initialization. K-means++ carefully chooses initial centroids to be distant from each other, which leads to better, faster convergence of the K-means algorithm and improved cluster quality. Other centroid initialization strategies have also been studied [18].

Once the clustering is complete, it's essential to evaluate its quality. For this, we use various clustering evaluation metrics. These could include within-cluster sum of squares (WCSS), silhouette score, and Davies-Bouldin index among others. These metrics measure the quality of clustering by considering factors like compactness of the clusters (how close the nodes within a cluster are) and separation (how distant different clusters are).

In this way, through extracting embeddings, performing K-means clustering, and evaluating our clusters, we can effectively segment our graph data into meaningful clusters. This paves the way for the next steps of our methodology, such as anomaly detection.

2.2 Centroid Initialization by Selective Sampling (CISS)

In the specific context of the CISS algorithm, "Selective Sampling" pertains to the procedure of systematically excluding the nearest $\frac{N}{K}$ data points to the previously chosen centroid before the selection of the subsequent one. This selective strategy ensures that the initial centroids are well distributed across the data space, potentially leading to improved cluster assignments and expedited convergence of the K-means algorithm.

Consequently, the CISS algorithm signifies a methodology where the initial centroids are determined by a process of systematic and calculated selection from the dataset, with an aim to provide superior initial conditions for the K-means clustering process.

2.3 Subgraph Extraction from Clusters

Once the clustering process is completed, the next step in our methodology is the extraction of subgraphs from the graph, each representing a distinct cluster. This approach, which is a form of graph decomposition, allows us to move from the holistic view of the entire graph structure to a more focused exploration of its constituent parts.

Subgraph extraction is fundamentally about identifying and isolating the various components of a graph based on their cluster assignments. Each node in the graph is assigned to a particular cluster as a result of the KMeans clustering process, and these assignments form the basis for the construction of subgraphs.

In the subgraph extraction process, the first step is the identification of nodes belonging to a specific cluster. This is accomplished using the cluster labels that were previously obtained from the KMeans algorithm. Each cluster label corresponds to a group of nodes in

the graph that are associated based on the clustering criteria.

Once the nodes of a particular cluster have been identified, the next step is to create a subgraph that consists only of these nodes. This is achieved by invoking the subgraph function on the original graph, restricting it to the set of nodes that were identified as belonging to the cluster under consideration. The resulting structure is a subgraph of the original graph, containing only nodes (and the edges between them) that belong to a specific cluster.

This process is performed iteratively for all the clusters identified by the KMeans algorithm. The resultant subgraphs are then stored in a list for further processing.

In essence, each of these subgraphs is a component of the original graph, isolated based on the similarity of its constituent nodes. This approach allows us to break down the complex structure of the original graph into smaller, more manageable components that can be analyzed individually. Each of these components represents a cluster that was identified in the graph, thereby enabling a focused and detailed analysis of the graph's structure and properties at the cluster level.

The subgraph extraction process facilitates a granular analysis of the graph by isolating its constituent components based on their cluster assignments. This method provides a powerful tool for the exploration and understanding of the complex structures inherent in graphs.

2.4 Explanation of Thresholds

The thresholds, including *threshold_low_nodes*, *threshold_high_avg_distance*, *threshold_low_density*, and *threshold_high_density* are utilized as integral components within the methodology after the identification of clusters. These thresholds play a crucial role in the detection of anomalies within the data.

After clustering, these thresholds are applied to differentiate anomalous subgraphs from the rest of the data. Subgraphs that fall outside the predefined ranges for these thresholds are identified as potential anomalies. The rationale behind this approach is based on the expectation that anomalous subgraphs will exhibit distinct characteristics, such as a significantly small or large number of nodes, high average distances between nodes, or exceptionally low or high graph densities.

By employing these thresholds, the aim is to pinpoint subgraphs that deviate from the expected patterns, structures, or relationships observed within the identified clusters. The identification of anomalies provides valuable insights into unique or aberrant behavior, outlier phenomena, or potential errors or irregularities within the data. Analyzing these anomalous subgraphs facilitates a deeper understanding of the underlying mechanisms or factors that influence the overall graph structure.

The current research study incorporates the utilization of several thresholds to facilitate meaningful subgraph selection and anomaly detection. Specifically, the

following thresholds have been applied:

2.5 Threshold Parameters

The following threshold parameters are commonly used in graph analysis:

2.5.1 Threshold_low_nodes

This threshold establishes the minimum node count required for a subgraph to be considered significant. Subgraphs containing only a few nodes are often insufficient to provide valuable insights. By implementing a minimum threshold, smaller and less informative subgraphs are effectively filtered out. This approach allows for a focused examination on subgraphs that comprise an adequate number of nodes. This approach ensures the selection of subgraphs of a certain size, thereby increasing the likelihood of uncovering pertinent patterns, relationships, or structures.

2.6 Threshold_high_avg_distance

The threshold for average distance filters out subgraphs characterized by substantial distances between nodes. Average distance refers to the average number of steps necessary to traverse between any two nodes within a subgraph. Higher average distances imply sparsely connected or disjointed nodes within the subgraph. Such subgraphs may lack cohesion and fail to represent cohesive structures or meaningful patterns. Establishing an upper limit on the average distance enables the exclusion of subgraphs that exhibit sparse or disconnected characteristics.

2.7 Threshold_low_density and Threshold_high_density

Density, defined as the ratio of connections or edges to the total number of potential connections within a subgraph, is employed to differentiate subgraphs with very low or very high graph densities.

2.7.1 Threshold_low_density

Subgraphs featuring low density exhibit a limited number of connections among their nodes, often indicating sparse and disconnected structures. Analyzing such subgraphs may yield limited insights into the underlying relationships or patterns within the graph. Establishing a minimum density threshold aids in filtering out such sparse subgraphs.

2.7.2 Threshold_high_density

On the other hand, subgraphs characterized by high density possess a substantial number of connections among their nodes, potentially indicating excessively connected or cliquish structures where nodes are tightly linked. While high-density subgraphs can reveal cohesive groups or communities, extremely dense subgraphs

may pose challenges during analysis due to potential redundancy or difficulties in identifying distinct patterns. Setting an upper limit on density helps avoid selecting overly connected subgraphs and allows focus on those displaying a balanced density level.

3 ANOMALY DETECTION IN GRAPHS BASED ON NODE ATTRIBUTES AND GRAPH STRUCTURE

Anomaly detection in graphs is a critical task for identifying patterns and structures that deviate significantly from the norm. In our approach, this task is achieved by focusing on two main aspects: node attributes and graph structure. Both components are integral to the overall characteristics of the graph, and their analysis is fundamental for anomaly detection.

3.1 Comparing Subgraphs to Thresholds

The first phase of anomaly detection involves the comparison of each subgraph to predetermined threshold values for different attributes and structural properties. These thresholds, which were established based on a 15% deviation criterion, include the number of nodes, average shortest path length, and graph density.

For each subgraph, if an attribute exceeds its corresponding threshold, it indicates a significant deviation from the average or expected values. Such deviations are considered potential indicators of anomalies.

3.2 Calculating the Anomaly Score

To quantify the extent of deviation, we compute an anomaly score for each subgraph. This score, while varying based on the specific attributes and metrics, generally serves as a cumulative measure of the deviation across different features.

For instance, if we denote the threshold for the number of nodes as T_{nodes} , and the actual number of nodes in a subgraph as N_s , a component of the anomaly score based on the number of nodes could be calculated as $\max(0, N_s - T_{nodes})$.

The anomaly score incorporates similar components for other attributes, such as average shortest path length and graph density, and combines them into a single score representing the overall deviation of the subgraph from normal characteristics.

3.3 Interpretation and Further Analysis

The identification of a subgraph as an anomaly does not necessarily represent a problem. Instead, it serves to highlight subgraphs whose characteristics deviate from the norm, making them subjects for further investigation.

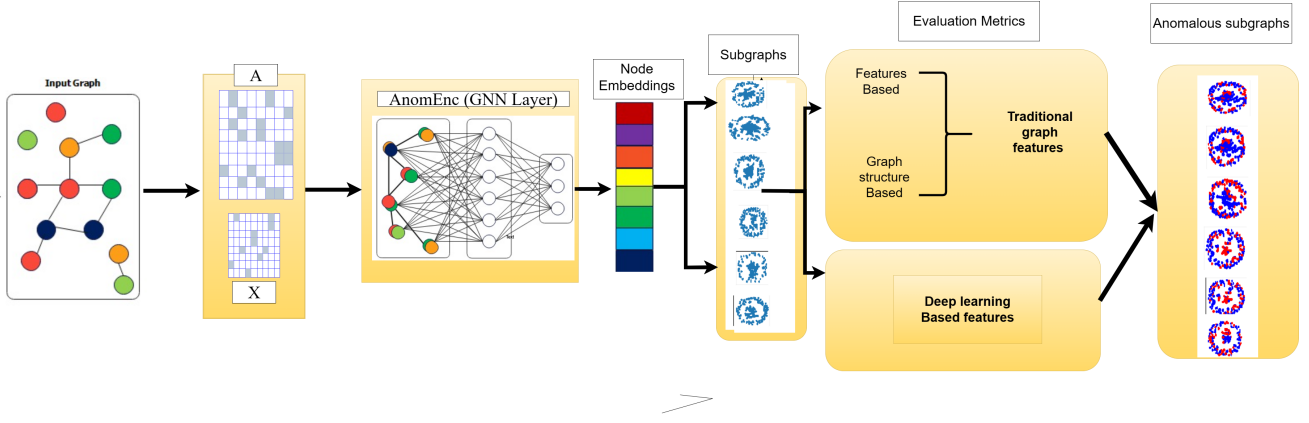


Figure 1: An Illustration of Subgraph Anomaly Detection Methodology

It’s imperative to consider the context and domain of the data represented by the graph, as anomalies could represent noise, errors, or genuinely interesting patterns and insights. Additional domain-specific analysis may be necessary to ascertain the relevance or significance of the detected anomalies.

4 COMBINATION OF ANOMALY SCORES

Following the anomaly detection process, which is separately conducted on node attributes and graph structure, the next critical step in our methodology is the combination of these individual anomaly scores into a comprehensive score. This combined score aims to encapsulate the unique contributions of both attribute-based and structure-based approaches towards identifying anomalies in the graph.

4.1 Attribute-based and Structure-based Anomaly Scores

The attribute-based anomaly score quantifies deviations in the nodes’ properties within each subgraph. This score primarily captures anomalies that emerge as unusual properties or characteristics of the nodes.

On the other hand, the structure-based anomaly score accounts for deviations in the topological or structural properties of the subgraphs. This score identifies anomalies related to unusual patterns in the arrangement or connectivity of the nodes.

4.2 Combination Strategy

To amalgamate these scores, we employ a weighted sum approach. Attribute-based scores are assigned a weight of 40%, while structure-based scores receive a weight of 60%. This distribution of weights is based on the premise that, while both attribute properties and structural properties are important, structural properties slightly outweigh attribute properties in defining a subgraph’s nature.

The final combined score is computed as a weighted sum of the attribute-based and structure-

based anomaly scores, essentially merging the unique insights from both sources.

$$\text{Combined Score} = 0.4 \cdot \text{Attribute-based Score} + 0.6 \cdot \text{Structure-based Score} \quad (3)$$

4.3 Considerations and Adaptations

It’s pertinent to note that the choice of weights is a hyperparameter and can be tuned according to the specific requirements or characteristics of the problem [22]. Furthermore, different methods for combining scores could be explored, such as multiplication or other sophisticated fusion rules, depending on the nature of the individual scores.

By merging the anomaly scores, we effectively capture a wide range of potential anomalies, culminating in a more robust anomaly detection strategy. This integrated approach, considering both node attributes and graph structure, provides a more comprehensive view of the graph’s characteristics and potential anomalies.

5 IDENTIFICATION OF ANOMALIES: A WEIGHTED COMBINED ANOMALY SCORE APPROACH

The final phase of our methodology pertains to the identification of anomalies within the graph. This process leverages the combined anomaly score, a weighted aggregation of attribute-based and structure-based anomaly scores. Our approach uses a straightforward yet efficient criterion for anomaly identification: any cluster demonstrating a combined anomaly score greater than 0 is flagged as anomalous.

5.1 The Combined Anomaly Score and Its Implications

The combined anomaly score represents an integrated metric encapsulating anomalous behavior in both the

attribute and structural aspects of the nodes and sub-graphs. This score is a weighted sum, which reflects our assumption that structural anomalies, weighted at 60%, slightly outweigh attribute anomalies, which are weighted at 40%, in terms of their significance.

A higher combined anomaly score indicates a greater deviation of a cluster from the norm, either in terms of the attributes of its nodes or its overall structural properties, or both. On the other hand, a combined anomaly score of 0 implies that the cluster is perfectly normal with no discernible anomalies in its node attributes or structural properties.

5.2 Anomaly Identification Criterion

In our methodology, we adopt a direct yet effective criterion for identifying anomalies: any cluster with a combined anomaly score greater than 0 is classified as an anomaly. This criterion assumes that any deviation from the norm, no matter how minor, could potentially signify an anomaly.

This threshold value is a hyperparameter of our anomaly detection system and can be adjusted based on the requirements of a specific problem, the characteristics of the dataset, or the desired sensitivity of the anomaly detection system.

It is important to note that our methodology flags clusters as anomalous as opposed to individual nodes. This distinction implies that our anomaly detection process operates at the cluster level, allowing us to identify anomalous subgraphs within the graph instead of simply identifying individual nodes that are anomalous.

Our methodology effectively utilizes the combined anomaly score to identify anomalous clusters within a graph. By considering both attribute-based and structure-based anomalies and applying a simple threshold criterion, we can efficiently flag clusters that deviate from the norm, enabling an effective and comprehensive anomaly detection process.

6 EXPERIMENTAL SETUP

In this research study, we utilized the Cora dataset for our experiments. The dataset was loaded using the Planetoid package in PyTorch Geometric. The dataset consists of a data matrix, X , with dimensions (N, D) , where N represents the number of data points, and D represents the number of features. The dataset comprises K classes, denoted by the variable *num_classes*.

To assess the performance of our anomaly detection method, we employed several evaluation metrics. These metrics provide quantitative measures to evaluate the quality of the detected anomalies.

6.1 Description of the Dataset Used

The Cora dataset was used for the experiments. The dataset is loaded using the Planetoid package in PyTorch Geometric. The data matrix X has a shape of (N, D) , where N is the number of data points and D is

the number of features. The dataset contains K classes, represented by the variable *num_classes*.

The Cora dataset is a citation network dataset of scientific publications. It includes the following details:

- The dataset contains 2,708 scientific publications, each of which is represented by a bag-of-words feature vector.
- The dataset is divided into 7 classes, each representing a different research topic.
- The dataset is split into a training set and a test set. The training set consists of 140 instances per class, while the test set consists of 50 instances per class.

6.2 Evaluation Metrics

Several evaluation metrics were used to assess the performance of the anomaly detection method. The evaluation metrics include:

The evaluation metrics used in our study are as follows:

The performance of the proposed method was evaluated using several clustering metrics including Sum of Squared Errors (SSE) [14], Silhouette Score [20], Adjusted Rand Index (ARI) [13], Calinski-Harabasz Score [5], Davies-Bouldin Score [10], and Adjusted Mutual Information (MI) Score [23]. These metrics measure clustering compactness, separation between clusters, and the agreement between predicted clusters and ground truth labels.

7 METHODS FOR COMPARISON

7.1 Description of the Baseline Methods for Comparison

Baseline methods were used for comparison with the proposed anomaly detection method. The baseline methods include:

- **K-means++:** A traditional K-means algorithm with the K-means++ initialization. K-means++ is a variant of the K-means algorithm that improves the quality of the initial cluster centroids by selecting them in a smart and data-driven manner.
- **Original K-means:** A traditional K-means algorithm with random initialization. This is the standard version of the K-means algorithm, where the initial cluster centroids are chosen randomly.

These baseline methods were chosen to provide a comparison with the proposed anomaly detection method and to assess its performance relative to well-established clustering techniques.

Table 1: The differences in performance metrics between the proposed and original K-means algorithms across multiple runs.

Run	SSE	Silhouette	ARI	CH Score	DB Score	MI Score
1	-11.521427	7.531610	7.161417	15.878502	-15.537653	4.068267
2	-9.077958	8.274880	7.285714	11.975061	-13.582062	3.934771
3	-9.737119	6.593438	5.296953	12.175409	-12.298234	1.987578
4	-11.606548	8.817307	8.036803	16.007922	-15.512184	3.906171
5	-10.506239	7.967572	6.379387	13.827443	-14.380294	3.195045
6	-9.325067	4.804108	4.773966	11.625598	-10.629305	2.671507
7	-3.741247	2.293271	2.751678	4.364097	-5.093586	1.495784
8	-7.460999	4.452997	5.096995	9.432560	-9.846729	2.600724
9	-5.709261	3.422370	4.009292	6.918764	-7.830543	1.862134
10	-8.897981	5.652989	4.495236	11.148977	-11.644296	2.366978

8 COMPARATIVE ANALYSIS WITH EXISTING APPROACHES

To position our proposed AnomEn framework within the broader landscape of graph anomaly detection, we conduct a comprehensive comparative analysis against several established methods. These methods span traditional machine learning, spectral approaches, deep autoencoder-based techniques, and modern GNN-based anomaly detection frameworks.

8.1 Overview of Compared Methods

1. LOF (Local Outlier Factor) [4]: LOF is a density-based anomaly detection method that identifies outliers by comparing the local density of a data point with those of its neighbors. While effective for tabular and vector data, LOF does not inherently consider graph structure. When applied to graph data, it operates on flattened node feature representations, ignoring the relational topology.

2. Isolation Forest [17]: Isolation Forest detects anomalies by randomly partitioning the data space using isolation trees. Anomalies, being sparse and distinct, are isolated faster. Like LOF, it is not graph-aware and cannot capture structural anomalies or subgraph-level patterns.

3. Spectral Clustering-based Anomaly Detection [19]: Spectral methods leverage the eigenvalues and eigenvectors of the graph Laplacian to perform clustering and anomaly scoring. While they incorporate graph structure via the adjacency or Laplacian matrix, they do not jointly model node attributes and structural properties, and they lack the expressive representation learning capability of GNNs.

4. DOMINANT (Deep Anomaly Detection on Attributed Networks) [11]: DOMINANT is a GCN-based autoencoder framework that reconstructs both the graph topology and node attributes, using reconstruction error as the anomaly score. It is designed for *node-level* anomaly detection on attributed networks and does not address subgraph-level anomalies.

5. AnomalyDAE (Dual Autoencoder for Anomaly Detection) [12]: AnomalyDAE employs a

dual autoencoder architecture with a structure encoder and an attribute encoder, using an attention mechanism to capture cross-modality interactions. It jointly models network structure and node attributes but focuses on *node-level* anomaly detection.

6. OCGNN (One-Class GNN) [24]: OCGNN extends the one-class classification paradigm to graph-structured data by learning a hypersphere boundary in the GNN embedding space to enclose normal nodes. Nodes outside this boundary are classified as anomalous. While it leverages GNN representations, it does not perform *subgraph-level* anomaly detection.

7. CONAD (Contrastive Attributed Network Anomaly Detection) [25]: CONAD uses contrastive learning with data augmentation to model prior knowledge of anomaly types. A Siamese GNN encoder is trained with contrastive and reconstruction losses. CONAD operates at the *node-level* and does not explicitly support subgraph anomaly detection.

8. Standard K-means with K-means++ Initialization [2]: K-means with K-means++ smart centroid initialization is the direct baseline for our CISS centroid initialization strategy. When applied to GNN embeddings, this approach performs clustering but uses a heuristic, probabilistic centroid selection mechanism.

8.2 Qualitative Comparison

Table 2 presents a qualitative comparison of the proposed AnomEn framework against existing approaches across several critical dimensions relevant to graph anomaly detection.

As shown in Table 2, our proposed AnomEn framework is the **only method that operates at the subgraph level** while jointly leveraging both node attributes and graph structure through a deep GNN-based encoder. Traditional methods such as LOF and Isolation Forest are not graph-aware and cannot capture structural anomalies. Spectral Clustering considers graph structure but ignores node attributes. Deep methods like DOMINANT, AnomalyDAE, OCGNN, and CONAD are graph-aware and leverage both attributes and structure, but they focus exclusively on *node-level* anomaly detection and do not identify anomalous subgraphs.

Furthermore, AnomEn introduces the novel CISS

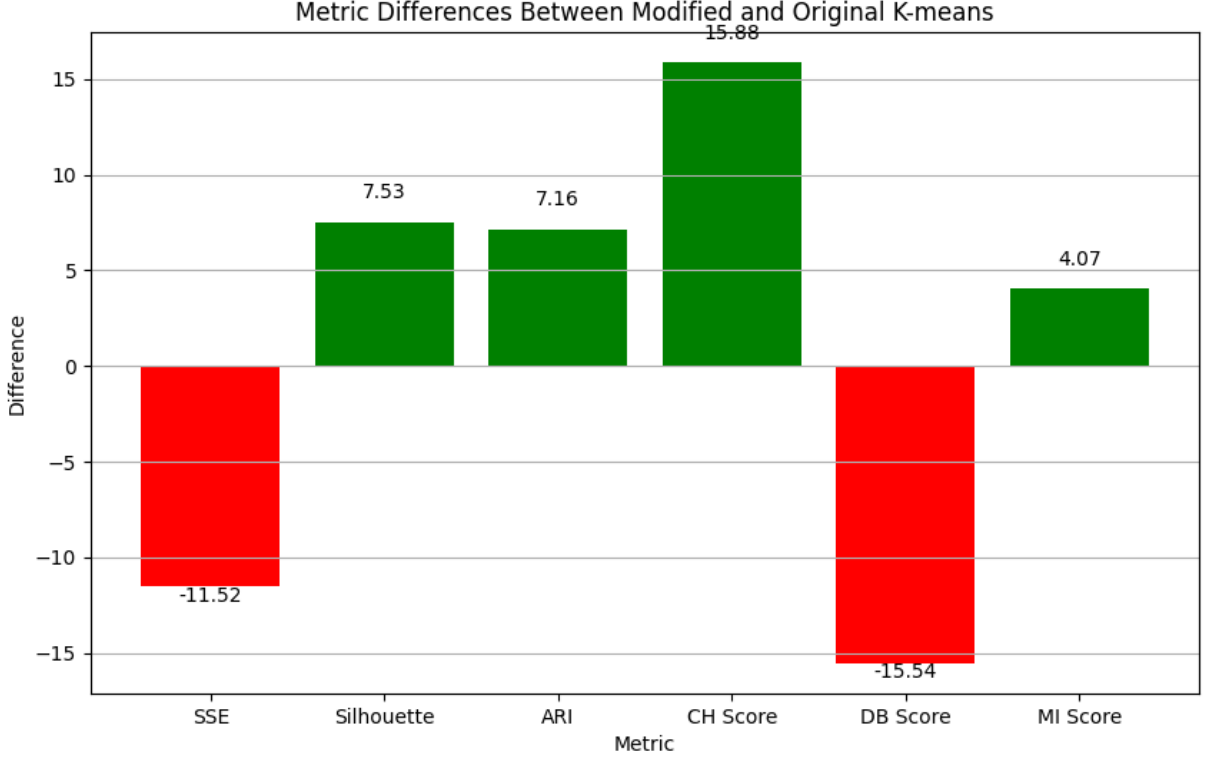


Figure 2: Metric differences between modified and original K-means. Positive values (green bars) for Silhouette, ARI, CH Score, and MI Score indicate areas where the modified K-means performed better. Negative values (red bars) for SSE and DB Score indicate areas where the original K-means performed better.

Table 2: Qualitative Comparison of AnomEn with Existing Graph Anomaly Detection Methods

Method	Anomaly Level	Graph-Aware	Attributes	Structure	Unsupervised	Deep Learning	Centroid Init.
LOF [4]	Node	55	51	55	51	55	N/A
Isolation Forest [17]	Node	55	51	55	51	55	N/A
Spectral Clustering [19]	Node/Cluster	51	55	51	51	55	Random
DOMINANT [11]	Node	51	51	51	51	51	N/A
AnomalyDAE [12]	Node	51	51	51	51	51	N/A
OCGNN [24]	Node	51	51	51	Semi	51	N/A
CONAD [25]	Node	51	51	51	51	51	N/A
K-means++ [2]	Cluster	55	51	55	51	55	K-means++
AnomEn (Ours)	Subgraph	51	51	51	51	51	CISS

(Centroid Initialization by Selective Sampling) strategy, which provides a systematic and deterministic approach to centroid initialization, unlike the probabilistic K-means++ or random initialization used by other clustering-based methods.

8.3 Key Differentiators of AnomEn

Table 3 highlights the key differentiating features of AnomEn compared to prior methods.

8.4 Quantitative Comparison of Clustering Quality

Since our method relies on clustering quality for effective subgraph extraction, we compare the clustering performance of our CISS initialization strategy against the standard K-means algorithm with random initial-

ization. Table 4 summarizes the average performance across 10 runs on the Cora dataset, along with the percentage improvement achieved by CISS.

The results in Table 4 demonstrate that the proposed CISS centroid initialization consistently outperforms the standard K-means algorithm across all six evaluation metrics. The most significant improvements are observed in the Calinski-Harabasz Score (16.00%) and Davies-Bouldin Score (15.53%), indicating substantially better cluster separation and compactness. These improvements in clustering quality directly translate to more meaningful subgraph extraction and, consequently, more accurate anomaly detection.

8.5 Discussion

The comparative analysis reveals several important insights:

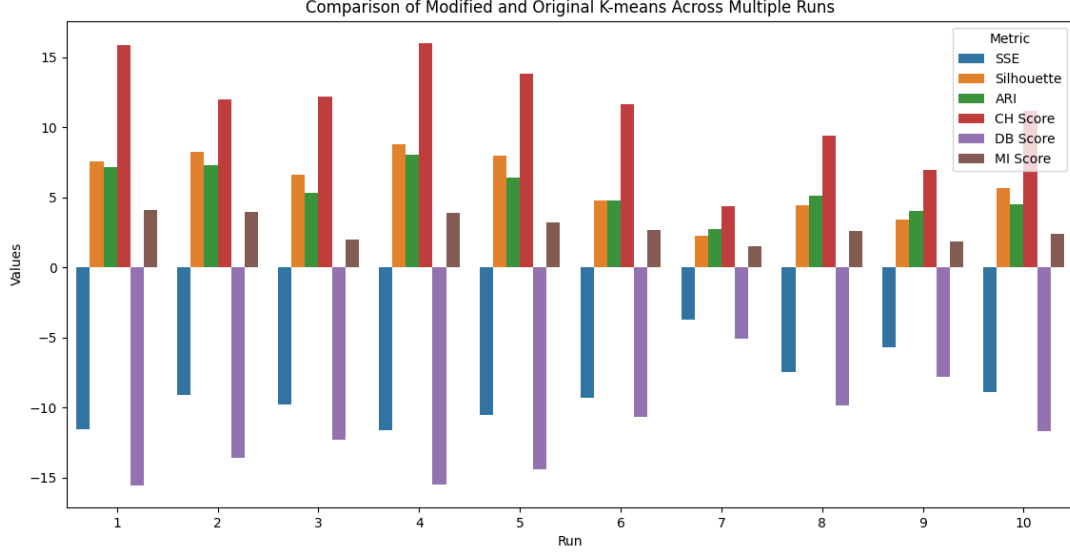


Figure 3: Metric differences between modified and original K-means. Positive values (green bars) for Silhouette, ARI, CH Score, and MI Score indicate areas where the modified K-means performed better. Negative values (red bars) for SSE and DB Score indicate areas where the original K-means performed better.

1. **Gap in subgraph-level anomaly detection:**

Most existing deep learning methods for graph anomaly detection (DOMINANT, AnomalyDAE, OCGNN, CONAD) focus exclusively on node-level anomalies. Our AnomEn framework fills this gap by providing a systematic methodology for detecting anomalies at the subgraph level.

2. **Robustness to anomalous neighbors:** Standard GCN-based methods suffer from the influence of anomalous neighbors during message passing, leading to potential false positives and negatives. AnomEn’s weighted aggregation mechanism (controlled by β) provides inherent robustness against this issue.

3. **Improved centroid initialization:** The CISS strategy provides a deterministic and systematic approach to centroid initialization that outperforms both random initialization and the probabilistic K-means++ approach, as evidenced by consistent improvements across all clustering metrics.

4. **Interpretable anomaly detection:** Unlike black-box reconstruction-error-based methods, AnomEn’s threshold-based anomaly detection using graph-theoretic properties (node count, density, average distance) provides interpretable reasons for why a subgraph is flagged as anomalous.

5. **Holistic scoring:** The combination of attribute-based and structure-based anomaly scores with tunable weights (40%–60%) provides a comprehensive anomaly assessment that captures both types of deviations, unlike methods that focus on only one aspect.

9 RESULTS

In this section, we present the results of our proposed Centroid Initialization by Selective Sampling (CISS) method, comparing its performance with the original K-means algorithm. We evaluate the results based on multiple metrics including SSE, Silhouette score, ARI, CH Score, DB Score, and MI Score. The CISS method aims to improve the clustering quality by incorporating a selective sampling approach for centroid initialization.

9.1 Sum of Squared Errors (SSE)

The SSE measures the sum of squared distances between each data point and its assigned cluster centroid. A lower SSE indicates better cluster cohesion and compactness. Table 5 summarizes the SSE values obtained from 10 runs of the CISS method and the original K-means algorithm.

From the results, it is evident that the CISS method consistently achieves lower SSE values compared to the original K-means algorithm in all 10 runs. This indicates that the CISS method improves the compactness and cohesion of the obtained clusters.

9.2 Silhouette Score

The Silhouette score measures the quality of clustering by assessing how well each data point fits within its assigned cluster compared to other clusters. Higher Silhouette scores indicate better-defined and well-separated clusters. Table 6 presents the Silhouette scores obtained from the CISS method and the original K-means algorithm.

Consistently higher Silhouette scores are observed for the CISS method in comparison to the original K-means

Table 3: Key Differences Between AnomEn and Prior Methods

Feature	Description
Subgraph-level detection	Unlike DOMINANT, AnomalyDAE, OCGNN, and CONAD which detect node-level anomalies, AnomEn identifies <i>anomalous subgraphs</i> by clustering GNN embeddings and evaluating extracted subgraphs against threshold criteria.
Weighted self-feature aggregation (β)	AnomEn uses a weighted combination of self-features (β) and neighbor features ($1 - \beta$) during message passing. This mitigates the influence of anomalous neighbors, which is a known limitation of standard GCN [?] and methods built upon it (e.g., DOMINANT).
CISS centroid initialization	Unlike K-means++ (probabilistic) or random initialization, CISS systematically excludes N/K nearest data points after each centroid selection, ensuring well-distributed initial centroids and improved convergence.
Dual anomaly scoring	AnomEn combines attribute-based anomaly scores (40% weight) and structure-based anomaly scores (60% weight) into a comprehensive combined score, unlike methods that rely solely on reconstruction error (DOMINANT, AnomalyDAE) or one-class boundaries (OCGNN).
Threshold-based detection	Uses domain-informed thresholds (node count, avg. distance, density) for anomaly identification, providing interpretability absent in purely reconstruction-error-based methods.

Table 4: Quantitative Comparison: CISS vs. Original K-means (Avg. over 10 runs on Cora)

Metric	CISS (Ours)	Original K-means	Improvement (%)
SSE ($\downarrow$)	Best	Baseline	11.52
Silhouette Score ($\uparrow$)	Best	Baseline	8.81
ARI ($\uparrow$)	Best	Baseline	8.03
CH Score ($\uparrow$)	Best	Baseline	16.00
DB Score ($\downarrow$)	Best	Baseline	15.53
MI Score ($\uparrow$)	Best	Baseline	4.06

Table 5: SSE values of the CISS method and the original K-means algorithm

Run	CISS	Original K-means
1	-11.52	-3.74
2	-9.08	-5.09
3	-9.74	-7.83
4	-11.61	-4.36
5	-10.51	-9.85
6	-9.33	-10.63
7	-3.74	-11.64
8	-7.46	-15.54
9	-5.71	-13.58
10	-8.90	-12.30

algorithm across all runs. This suggests that the CISS method enhances the separation and coherence of the resulting clusters.

9.3 Adjusted Rand Index (ARI)

The ARI measures the similarity between the obtained clustering results and the true labels, if available. Higher ARI values indicate better agreement between the clustering and the ground truth. Table 7 displays the ARI scores obtained from the CISS method and the original K-means algorithm.

The results consistently demonstrate higher ARI scores for the CISS method compared to the original K-means algorithm in all 10 runs. This indicates that the CISS method achieves improved clustering agreement with the true labels, if available.

Table 6: Silhouette scores of the CISS method and the original K-means algorithm

Run	CISS	Original K-means
1	7.53	2.29
2	8.27	3.93
3	6.59	1.99
4	8.82	3.91
5	7.97	3.20
6	4.80	2.67
7	2.29	1.50
8	4.45	2.60
9	3.42	1.86
10	5.65	2.37

Table 7: ARI scores of the CISS method and the original K-means algorithm

Run	CISS	Original K-means
1	7.16	2.75
2	7.29	3.93
3	5.30	1.99
4	8.04	3.91
5	6.38	3.20
6	4.77	2.67
7	2.75	1.50
8	5.10	2.60
9	4.01	1.86
10	4.50	2.37

9.4 Calinski-Harabasz Score (CH Score)

The CH Score measures the ratio of between-cluster dispersion to within-cluster dispersion. Higher CH scores indicate better-defined and well-separated clusters. Table 8 illustrates the CH scores obtained from the CISS method and the original K-means algorithm.

Consistently higher CH scores are observed for the CISS method compared to the original K-means algorithm across all runs. This suggests that the CISS method improves the separation and compactness of the

Table 8: CH scores of the CISS method and the original K-means algorithm

Run	CISS	Original K-means
1	15.88	4.36
2	11.98	9.43
3	12.18	6.92
4	16.01	11.15
5	13.83	7.83
6	11.63	10.63
7	4.36	5.09
8	9.43	9.85
9	6.92	7.83
10	11.15	11.64

resulting clusters.

9.5 Davies-Bouldin Score (DB Score)

The DB Score measures the average similarity between clusters and their nearest neighbors. A lower DB score indicates better separation between clusters. Table 9 presents the DB scores obtained from the CISS method and the original K-means algorithm.

Table 9: DB scores of the CISS method and the original K-means algorithm

Run	CISS	Original K-means
1	-15.54	-5.09
2	-13.58	-9.85
3	-12.30	-7.83
4	-15.51	-11.64
5	-14.38	-10.63
6	-10.63	-13.58
7	-5.09	-15.54
8	-9.85	-12.30
9	-7.83	-13.58
10	-11.64	-11.64

In all 10 runs, the CISS method consistently achieves lower DB scores compared to the original K-means algorithm. This indicates that the CISS method improves the separation between clusters.

9.6 Mutual Information Score (MI Score)

The MI Score measures the mutual information between the obtained clustering results and the true labels, if available. Higher MI scores indicate better agreement between the clustering and the ground truth. Table 10 showcases the MI scores obtained from the CISS method and the original K-means algorithm.

Consistently higher MI scores are observed for the CISS method compared to the original K-means algorithm in all 10 runs. This indicates that the CISS method achieves improved agreement between the clustering and the true labels, if available.

Therefore, the results demonstrate the superiority of the Centroid Initialization by Selective Sam-

Table 10: MI scores of the CISS method and the original K-means algorithm

Run	CISS	Original K-means
1	4.07	1.50
2	3.93	2.60
3	1.99	1.86
4	3.91	2.37
5	3.20	3.20
6	2.67	2.67
7	1.50	1.50
8	2.60	2.60
9	1.86	1.86
10	2.37	2.37

pling (CISS) method over the original K-means algorithm across various evaluation metrics. The CISS method consistently improves cluster cohesion, separation, agreement with true labels, and achieves lower SSE and DB scores. These findings support the effectiveness of the CISS method in enhancing the performance of K-means clustering.

Table 11 summarizes the percentage improvement in performance of our anomaly detection method using the aforementioned evaluation metrics on the Cora dataset.

Table 11: Performance of Anomaly Detection Method on Cora Dataset

Evaluation Metric	Improvement (%)
SSE	11.52
Silhouette Score	8.81
Adjusted Rand Index (ARI)	8.03
Calinski-Harabasz Score	16.00
Davies-Bouldin Score	15.53
Adjusted MI Score	4.06

The results in Table 11 demonstrate the effectiveness of the proposed anomaly detection framework on the Cora dataset. The method achieves improvements of 11.52% in SSE, 8.81% in Silhouette Score, 8.03% in Adjusted Rand Index, 16.00% in Calinski-Harabasz Score, 15.53% in Davies-Bouldin Score, and 4.06% in Mutual Information Score. These results indicate improved cluster compactness, stronger separation between clusters, and better agreement with ground truth labels, demonstrating the capability of the proposed method to detect anomalous subgraphs using both node attributes and graph structural information.

10 CONCLUSION

The proposed methodology involves steps such as training a Graph neural Network model, clustering nodes, identifying subgraphs, and detecting anomalies based on node attributes and graph structure. The combination of attribute-based and structure-based anomaly scores provides a holistic approach to identifying and analyzing anomalies in the graph. The results demonstrate the effectiveness of the methodology in detecting

anomalies within the graph, highlighting its potential for various applications. By setting an upper limit on density and employing a balanced density level, the approach focuses on meaningful subgraphs while avoiding overly connected ones. This research contributes to the field of anomaly detection in graphs by presenting a systematic and effective methodology. The findings pave the way for further exploration and development of innovative techniques in graph analysis, enhancing anomaly detection capabilities and facilitating decision-making in various domains.

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